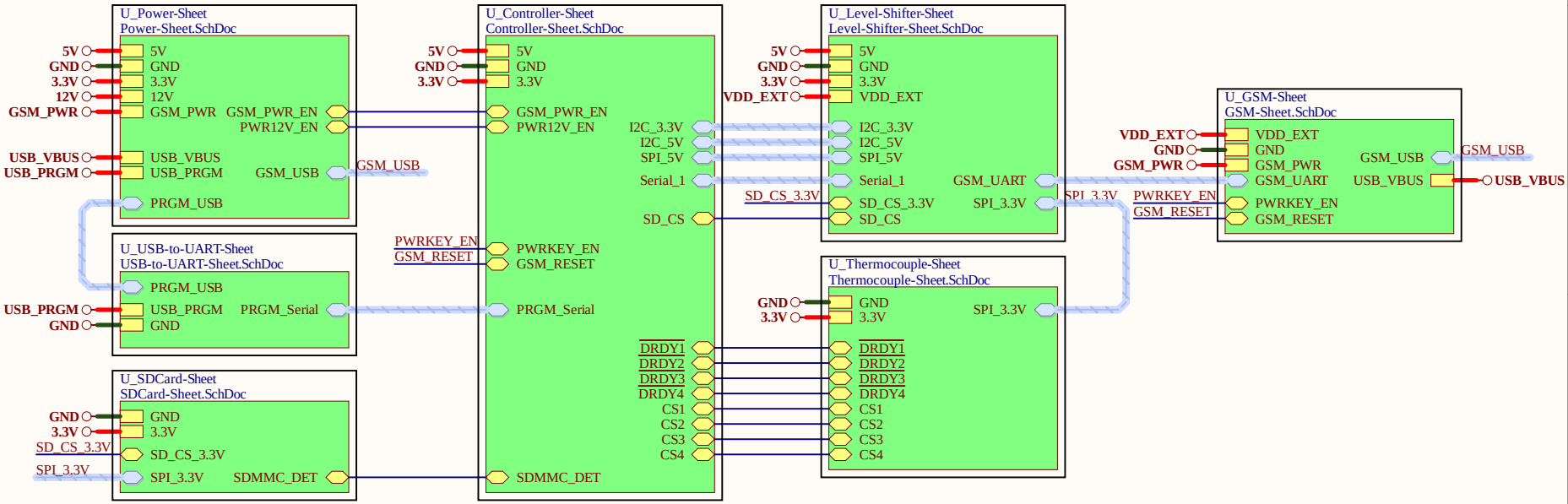
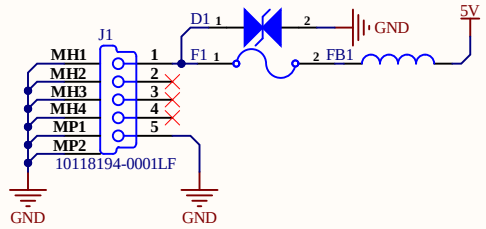


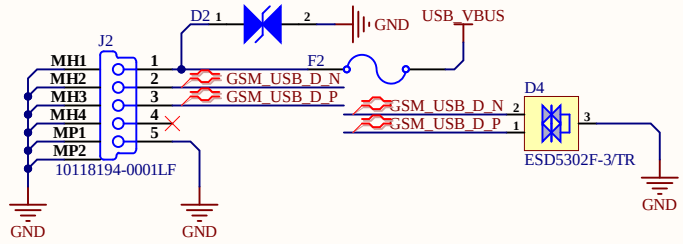
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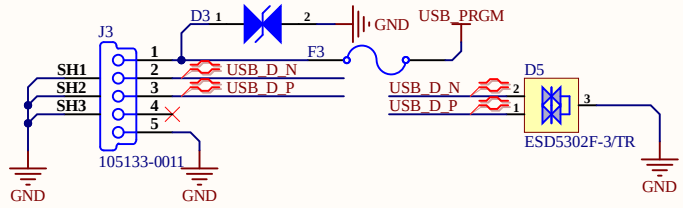
USB to Power Board



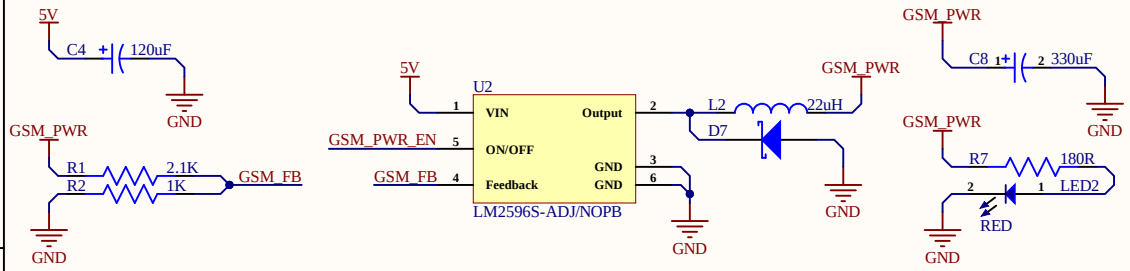
USB for GSM Updates



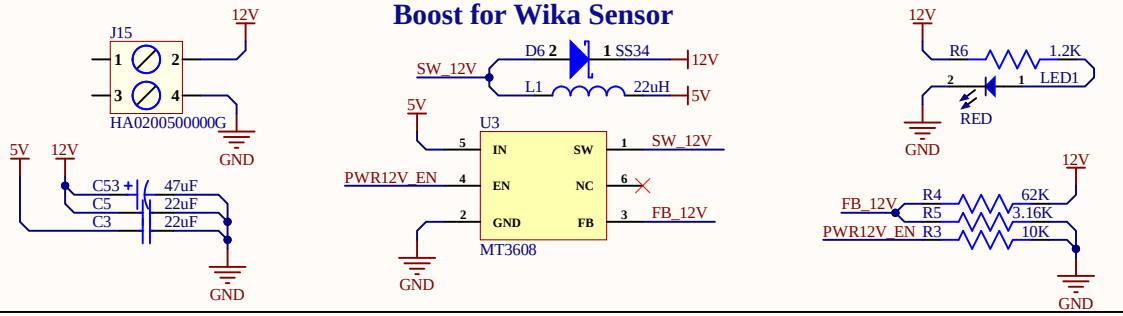
USB for Programming Mega2560



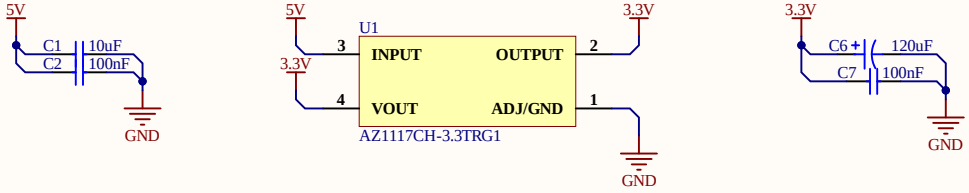
Buck for GSM Power



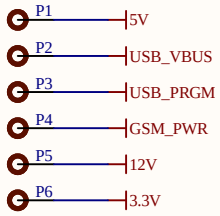
Boost for Wika Sensor

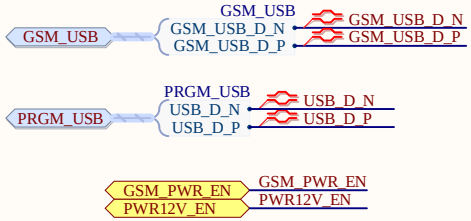


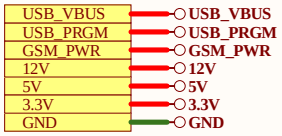
LDO 3.3V



Test Points



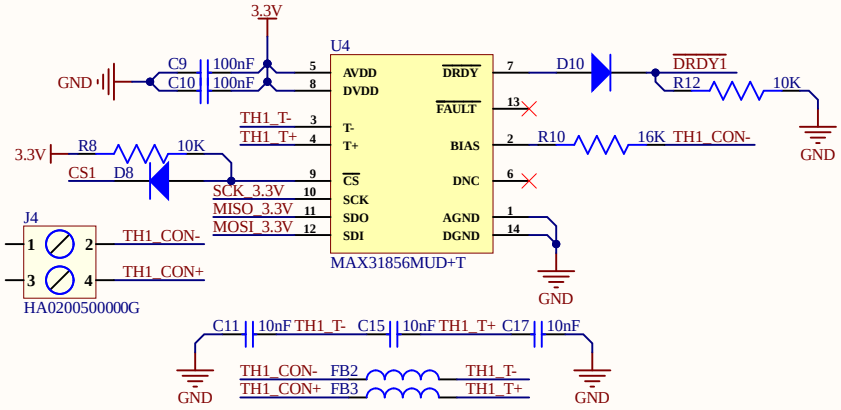




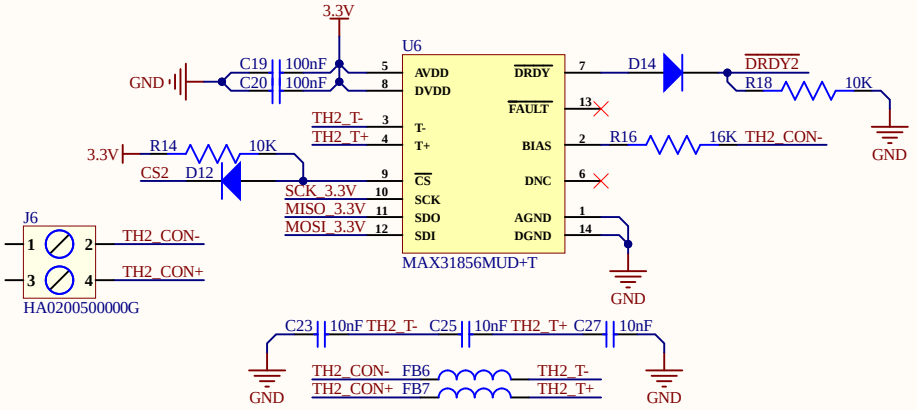
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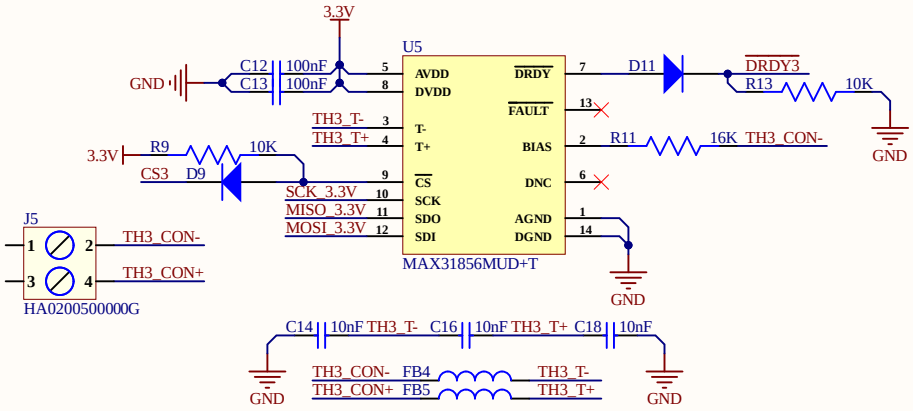
Thermocouple Input - 1



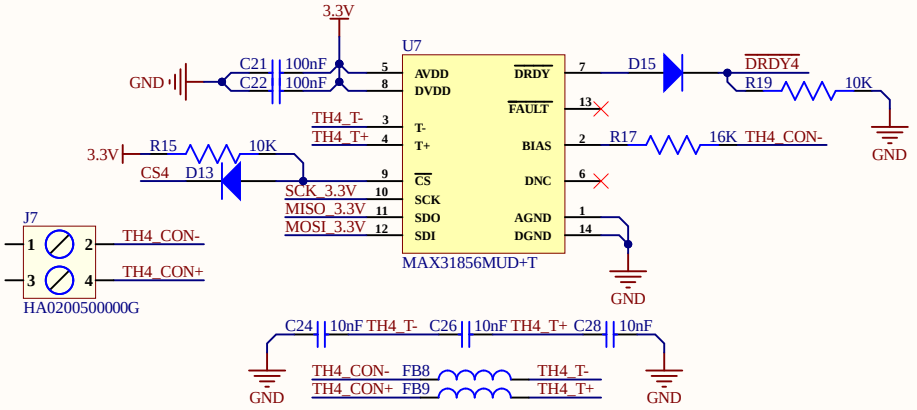
Thermocouple Input - 2



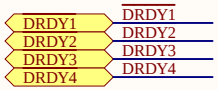
Thermocouple Input - 3

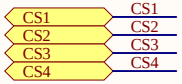


Thermocouple Input - 4







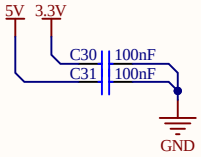


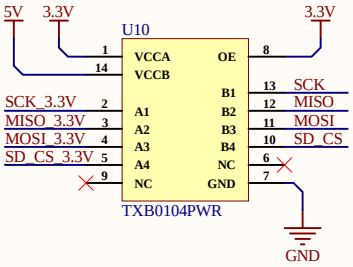


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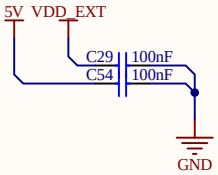
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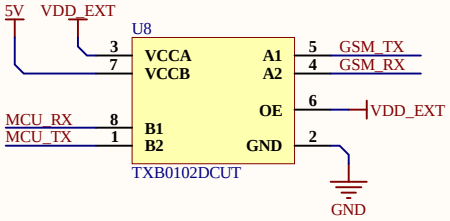
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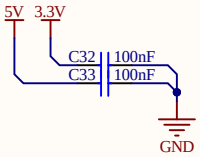


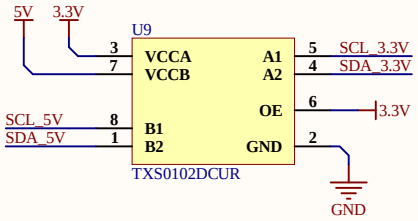
Level Shifter for Serial

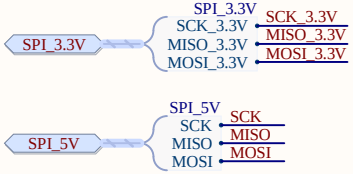


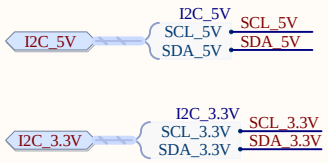


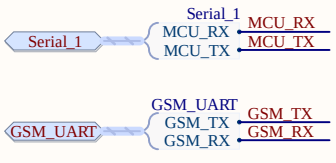
Level Shifter for I2C

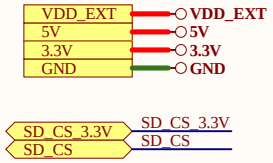






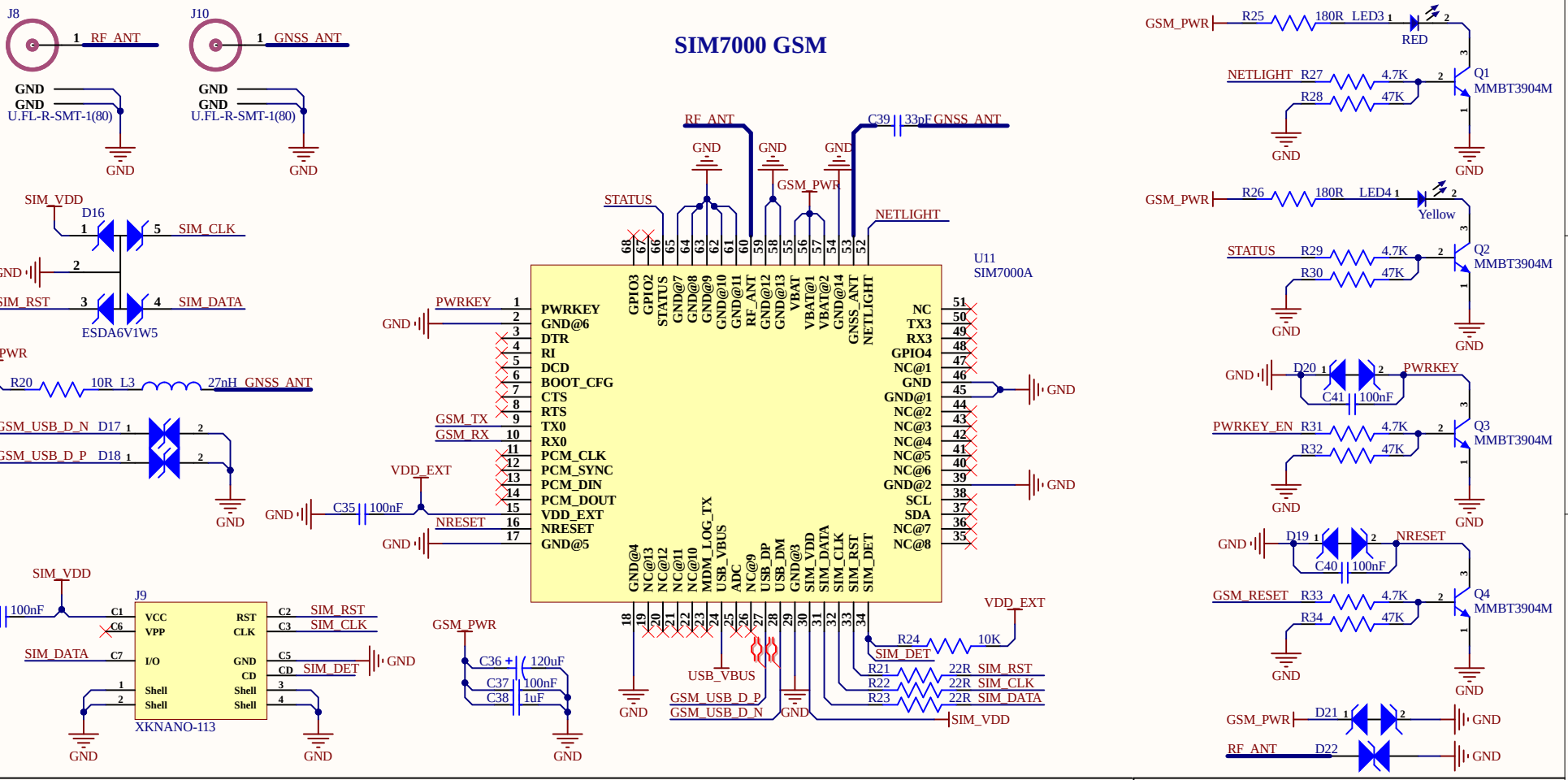


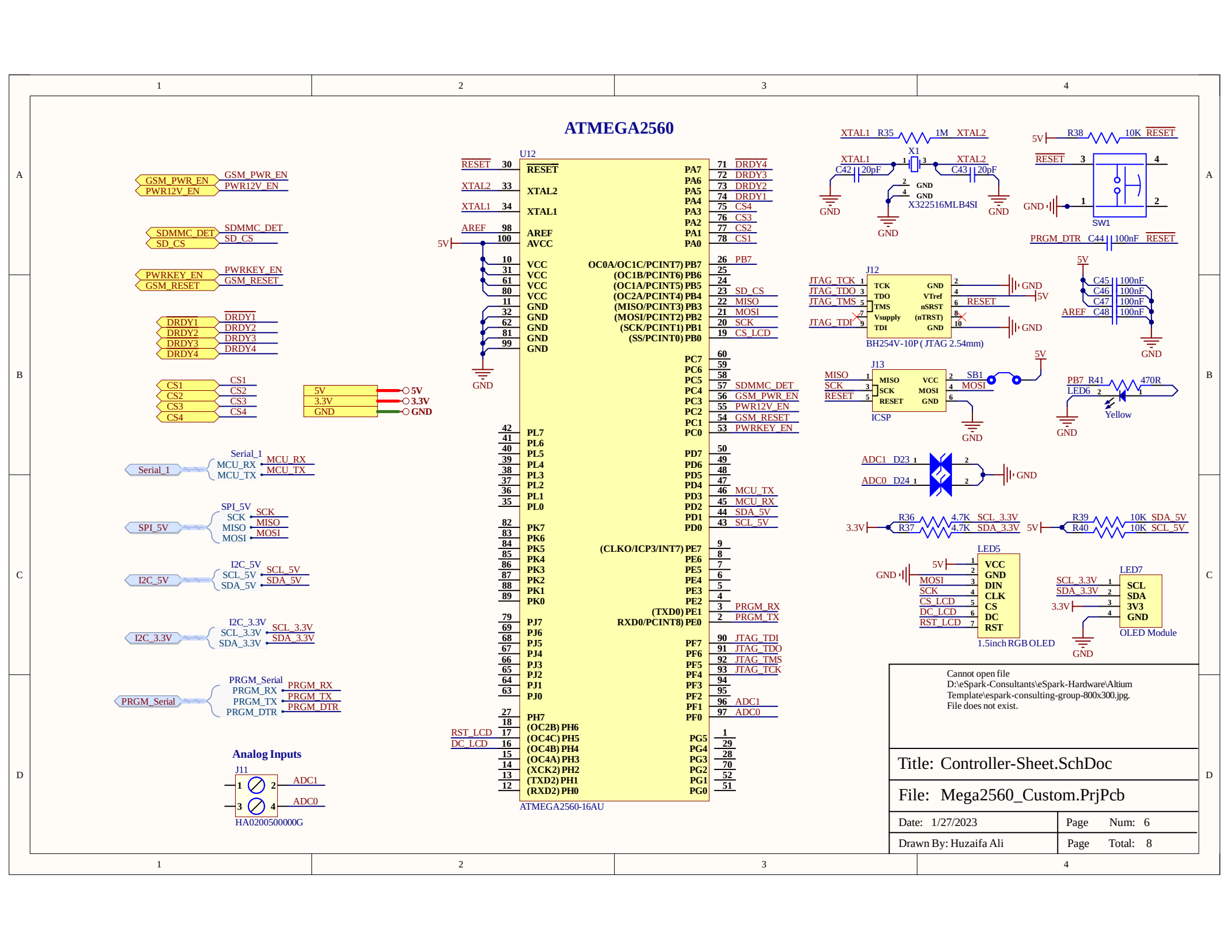




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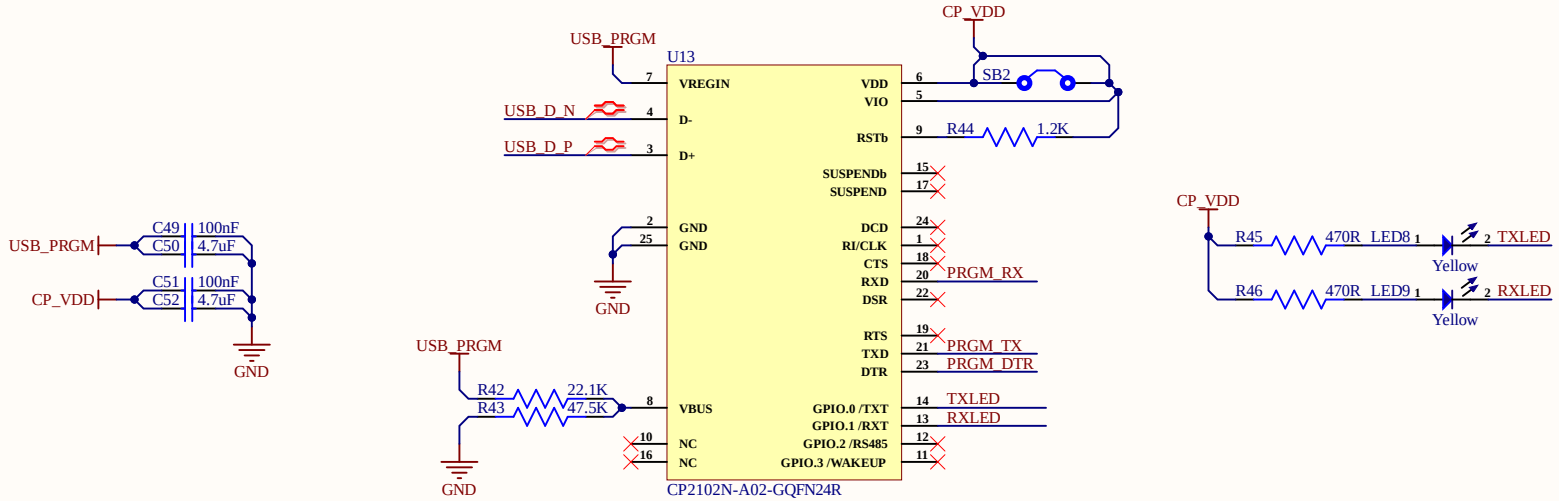

SIM7000 GSM
J8
J10
RF_ANT
GNSS_ANT
GND
U.FL-R-SMT-1(80)
SIM_VDD
D16
SIM_CLK
SIM_RST
SIM_DATA
ESDA6V1W5
GSM_PWR
R20
10R
L3
27nH_GNSS_ANT
GSM_USB_D_N
D17
GSM_USB_D_P
D18
J9
XKNANO-113
C34
100nF
C35
C36
120uF
C37
C38
1uF
VDD_EXT
NRESET
PWRKEY
STATUS
NETLIGHT
GSM_TX
GSM_RX
C39
33pF
U11
SIM7000A
PWRKEY
GND@6
DTR
RI
DCD
BOOT_CFG
CTS
RTS
TX0
RX0
PCM_CLK
PCM_SYNC
PCM_DIN
PCM_DOUT
VDD_EXT
NRESET
GND@5
GPIO3
GPIO2
STATUS
GND@7
GND@8
GND@9
GND@10
GND@11
RF_ANT
GND@12
GND@13
VBAT
VBAT@1
VBAT@2
GND@14
GNSS_ANT
NETLIGHT
NC
TX3
RX3
GPIO4
NC@1
GND
GND@1
NC@2
NC@3
NC@4
NC@5
NC@6
GND@2
SCL
SDA
NC@7
NC@8
GND@4
NC@13
NC@12
NC@11
NC@10
MDM_LOG_TX
USB_VBUS
ADC
NC@9
USB_DP
USB_DM
GND@3
SIM_VDD
SIM_DATA
SIM_CLK
SIM_RST
SIM_DET
R24
10K
R21
22R
R22
R23
SIM_RST
SIM_CLK
SIM_DATA
SIM_DET
R25
180R
LED3
RED
R27
4.7K
R28
47K
Q1
MMBT3904M
R26
LED4
Yellow
R29
R30
Q2
R31
R32
Q3
D20
C41
D19
C40
R33
R34
Q4
D21
D22
GSM_UART
GSM_USB
PWRKEY_EN
GSM_RESET
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Date: 1/27/2023
Page Num: 5
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Page Total: 8



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Date: 1/27/2023	Page Num: 6
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USB-to-UART for Programming





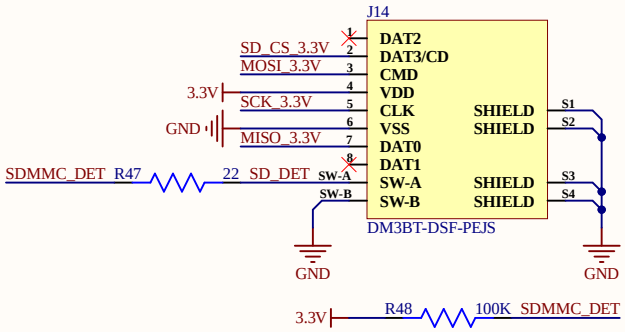


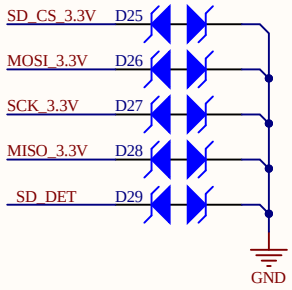


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SD Card Interface





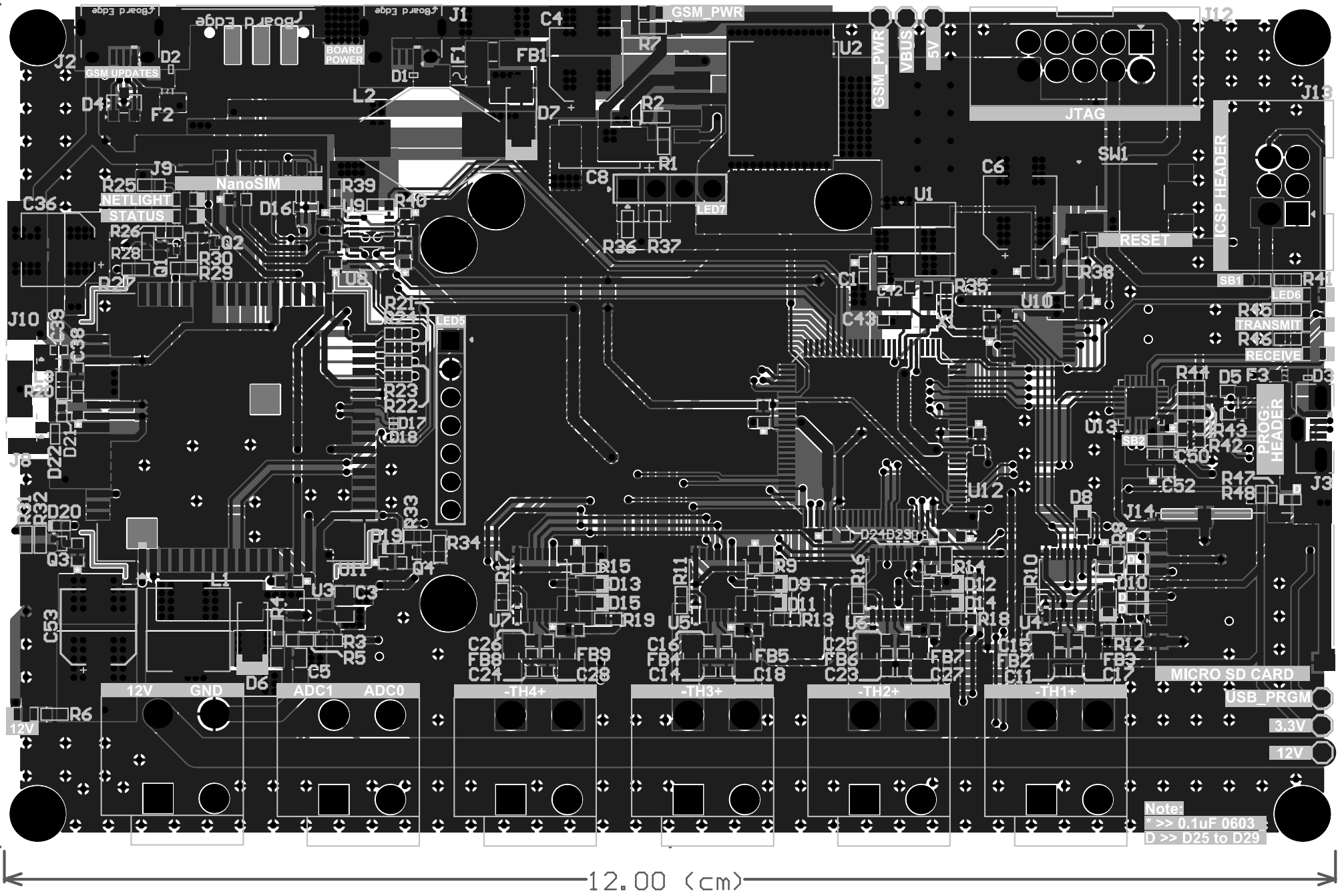






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Drawn By: Huzaifa Ali	Page Total: 8


7.60 (cm)
12.00 (cm)
GSM_PWR
VBUS
5V
JTAG
ICSP HEADER
RESET
NanoSIM
NETLIGHT
STATUS
GSM UPDATES
BOARD POWER
TRANSMIT
RECEIVE
PROG. HEADER
MICRO SD CARD
USB_PRGM
3.3V
12V
GND
ADC1
ADC0
-TH4+
-TH3+
-TH2+
-TH1+
Note:
* >> 0.1uF 0603
D >> D25 to D29